

# Equivalent Width Diagnostics of Solar Magnetic Activity

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## ABSTRACT

Radial velocity detections of true Earth analogs require precision near 10 cm/s, yet magnetic activity on Sun-like stars, including starspots, faculae, and suppressed convective blueshift, can produce apparent velocity shifts that are comparable to or larger than the planetary signal. This study investigates whether variability in a magnetically sensitive Fe I line in high-resolution, near-infrared solar spectra is consistent with a solar signal and distinguishable from residual telluric contamination, and whether similar near-infrared observations in Sun-like stars can help mitigate stellar noise in radial velocity measurements. Using a solar tracker and a high-resolution near-infrared spectrograph, daily solar spectra spanning over 10 months were obtained and analyzed, with the periodogram analysis focused on a 63-day subset and on a strongly Zeeman-sensitive Fe I absorption line near 1565 nm as a useful tracer of activity-related changes in the solar photosphere. To test whether the observed variability could arise from Earth's atmosphere rather than the solar photosphere, a nearby telluric H<sub>2</sub>O absorption feature was used as a control. Changes in the Fe I line's equivalent width were measured, and periodogram analysis was applied to search for activity-related periodic signals. After accounting for instrumental and atmospheric effects, evidence for activity-driven equivalent width variations was found in the form of a prominent peak near the solar rotation period. These results support the use of near-infrared, magnetically sensitive line diagnostics as activity indicators that can help separate stellar variability from true planetary Doppler shifts, improving the reliability of exoplanet detections with radial velocity surveys.

## INTRODUCTION

Finding Earth-mass planets that orbit Sun-like stars is a major step toward identifying potentially-habitable planets and understanding how common it is for planetary systems to be analogous to the solar system. These discoveries help prioritize promising targets for future observations that can probe planetary atmospheric conditions and habitability. Radial Velocity (RV) measures the tiny back-and-forth motion of a star along our line of sight as it responds to the gravitational pull of an orbiting planet. This motion produces periodic shifts in stellar absorption lines via the Doppler effect: motion towards the observer shifts lines to have shorter wavelengths, while motion away shifts them to have longer wavelengths. These phenomena are known as blueshifts and redshifts respectively. Identifying systems with Earth-analogs requires extremely precise RV measurements, because an Earth analog only induces Doppler shifts of only about 10 cm s<sup>-1</sup> in a Sun-like star (1).

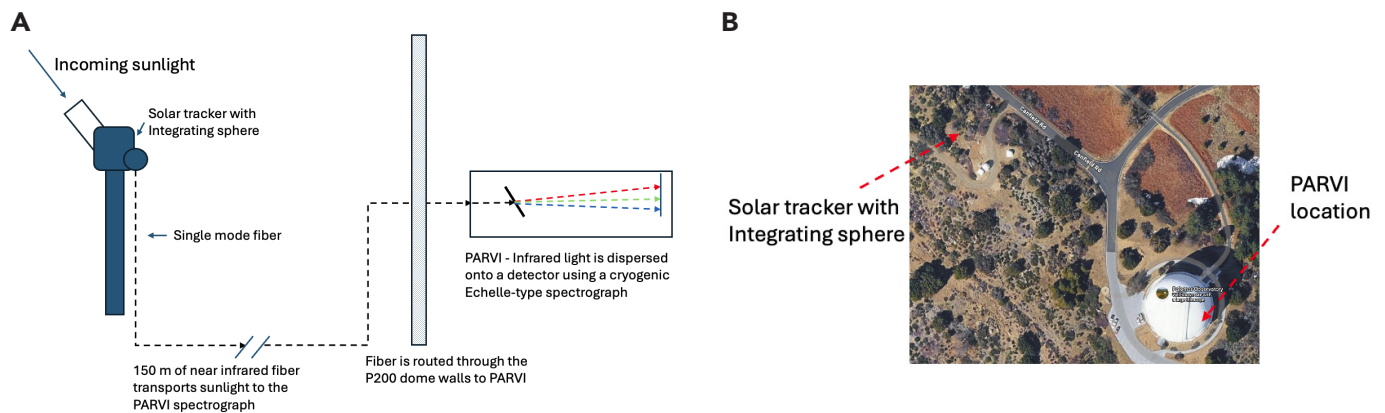
A major obstacle to achieving this precision is stellar jitter, which causes time-varying apparent velocity shifts driven by physical processes such as convection, granulation, and magnetic activity on the stellar surface (1). These processes distort absorption lines, which are the dark features in a spectrum formed when atoms in the stellar atmosphere absorb light at specific wavelengths. Because absorption lines anchor RV measurements, time-dependent changes in their depths and shapes introduce noise that can mimic or obscure planetary Doppler signals (2).

Since the size and character of stellar jitter can depend on wavelength, comparing measurements across wavelength ranges can help separate solar activity signals, and more generally stellar activity signals, from true Keplerian motion (3). The near-infrared (NIR) is therefore a useful regime for studying wavelength-dependent stellar activity signals. Prior work has shown that magnetic activity can influence RV measurements

differently across wavelength ranges, including in the NIR (3). NIR solar studies have also shown that this wavelength range contains rich diagnostic information about magnetic fields, atmospheric structure, and line formation (4). The NIR spans roughly 10,000–4,000 cm<sup>-1</sup> (1–2.5 μm), which is beyond the red end of the visible spectrum (4). Many NIR spectral lines form deeper in the stellar photosphere, the visible surface from which most photons escape, than their optical, or visible-light, counterparts (4). If the physical conditions and magnetic structuring in these deeper layers differ from those higher in the photosphere, then NIR lines may respond differently to activity than optical lines. This possibility suggests that combining NIR and optical diagnostics could provide an additional way to test whether wavelength-dependent behavior helps separate stellar activity from true Keplerian motion (4).

The Sun offers a uniquely accessible test case for evaluating this approach because it is the only star whose surface can be directly imaged. As a result, disk-integrated “Sun-as-a-star” signals that are obtained by summing light over the full solar disk can be related to the resolved surface features and magnetic structures that drive the observed variability. The Southern California Solar Calibrator (SoCalSoCal) solar monitoring program feeds disk-integrated sunlight into Palomar Radial Velocity Instrument (PARVI), a high-resolution NIR spectrograph at Palomar Observatory, to collect daily, high signal-to-noise solar spectra (Figure 1). These observations complement optical solar RV datasets from instruments such as NN-EXPLORE Exoplanet Investigations with Doppler spectroscopy (NEID) and High Accuracy Radial velocity Planet Searcher for the Northern hemisphere (HARPS-N), enabling controlled multiwavelength comparisons of activity driven signals (5–8).

This work focuses on the H-band spectral region around 6410.26 cm<sup>-1</sup>, a region with relatively low telluric contamination from atmospheric absorption compared to many neighboring NIR intervals. Telluric



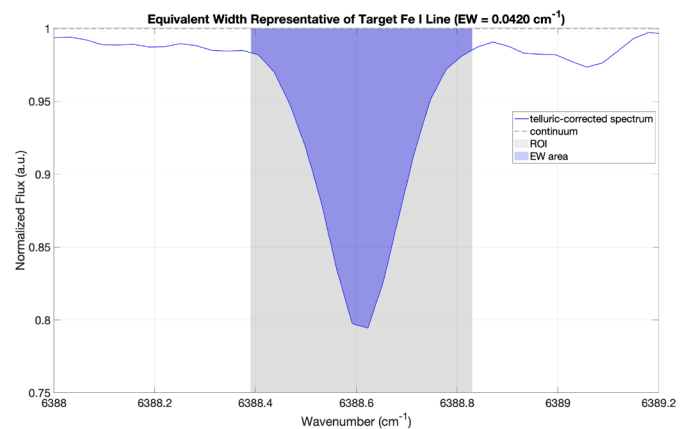
**Figure 1: Southern California Solar Calibrator (SoCalSoCal) to Palomar Radial Velocity Instrument (PARVI) optical path for Sun-as-a-star observations.** (A) Schematic of the observing setup. The SoCalSoCal solar tracker collects sunlight with a 75 mm telescope and feeds the light into an integrating sphere to spatially scramble the solar disk. The integrating sphere homogenizes the light, producing a more uniform Sun-as-a-star signal for repeatability spectroscopic measurements. Light is then coupled into ~150 m of single-mode fiber (SMF) and transported to the PARVI spectrograph in the Palomar Hale Telescope dome, where the resulting high-resolution solar spectra are recorded for later analysis. (B) The Google Maps inset shows the locations of the SoCalSoCal tracker and PARVI.

contamination refers to absorption features imprinted by Earth's atmosphere, especially water vapor, that overlap the stellar spectrum and can bias measurements if not corrected. The H-band is part of the NIR window, spanning  $5555.56$  to  $6666.67\text{ cm}^{-1}$ , where atmospheric transmission is relatively high because Earth's atmosphere absorbs less strongly than in many neighboring wavelength ranges, enabling high-quality ground-based observations. Within this region, H-band spectra include a particularly magnetically sensitive Fe I line at  $6388.6\text{ cm}^{-1}$  ( $1564.8\text{ nm}$ ) with a large effective Landé  $g$ -factor ( $g_{\text{eff}} = 3$ ) (4, 9). The Landé  $g$ -factor reveals how strongly a line undergoes Zeeman splitting, the separation of atomic energy levels in the presence of a magnetic field, making this Fe I line a useful tracer of activity-related changes in the solar photosphere and a strong candidate for testing whether the observed variability is solar in origin rather than telluric.

In this study, the equivalent width (EW) of the Fe I  $6388.6\text{ cm}^{-1}$  line (Figure 2) is monitored. EW quantifies absorption-line strength as the width of a rectangle at the continuum level with the same area as the absorption features, reflecting combined changes in line depth and width. Because it captures overall line strength without requiring detailed modeling of the line profile, EW provides a simple way to track variability in the feature. Although SoCalSoCal spectra were collected over a longer observing campaign, the period analysis in this paper focuses on a contiguous 63-day interval with minimal gaps to reduce aliases caused by the observational sampling pattern, or window function, in the periodograms (10).

This study investigates whether the EW of the magnetically sensitive Fe I  $6388.6\text{ cm}^{-1}$  line varies on solar rotation timescales and examines whether that variability can be distinguished from residual telluric systematics. To test for residual telluric effects, periodogram peaks in an  $\text{H}_2\text{O}$  control line were also tracked. Because this line originates from Earth's atmosphere, it should vary with terrestrial observing conditions, such as humidity and air mass, rather than with the Sun's rotation. A strong periodogram peak near the solar rotation period would suggest telluric or analysis-related contamination, whereas the absence of such a peak increases confidence that the Fe I rotation-period signal is solar in origin. This can be tested by measuring EWs in micro-telluric-corrected spectra and using Lomb–Scargle periodograms to search for signals near the Sun's rotation timescale and other activity-related timescales. By connecting these NIR EW variations to contemporaneous

optical diagnostics, this work evaluates whether NIR activity indicators can complement optical measurements in extreme-precision RV (EPRV) surveys aimed at detecting Earth-like planets around Sun-like stars.



**Figure 2: Representative equivalent-width (EW) measurement for the Fe I  $6388.6\text{ cm}^{-1}$  absorption line.** The telluric-corrected, continuum-normalized solar spectrum (blue line) is shown over the region of interest (ROI), where ROI denotes the wavenumber window used to compute EW. For this example exposure, the EW ROI spans  $[\omega_1, \omega_2] = [6388.39, 6388.83]\text{ cm}^{-1}$  (vertical bounds of the shaded region). Within the ROI, the normalized flux ranges from  $F_{\text{min}} = 0.794$  at  $\omega_{\text{min}} = 6388.62\text{ cm}^{-1}$  to  $F \approx 1.00$  in the local continuum. The blue shaded area marks the integrated EW of the absorption feature relative to the local continuum. Additional weak structures outside the shaded ROI (for example, near  $\sim 6389.05\text{ cm}^{-1}$ ) is not included in the EW integration.

## METHODS

### Observations

Solar spectra were acquired using SoCalSoCal, which is equipped with a 75-mm telescope that directs sunlight into an integrating sphere—a reflective spherical cavity that mixes light so the output is uniform. This disk-integration is essential because it blends light from the entire visible Sun into a single spectrum, like a distant star. Any measured

variability, therefore, reflects changes in the overall coverage of features such as sunspots and faculae across the solar disk rather than changes confined to one small region. An optical fiber carries the near-infrared (NIR) sunlight from the sphere to the PARVI spectrograph, located about 150 m away (Figure 1). On clear, cloud-free days with stable atmospheric transparency, SoCalSoCal automatically tracks the Sun, and the resulting data acquisition consists of a series of spectra with integration times of approximately 186 s. On such days, SoCalSoCal yields about 200 high-quality spectra per day.

PARVI is a diffraction-limited, high-resolution NIR echelle spectrograph commissioned at Palomar Observatory (Palomar Mountain, CA) with spectral coverage from 1.1 to 1.76  $\mu\text{m}$ . It is connected by single-mode fiber to the 200-inch Hale telescope at Palomar Observatory and can also be fed into SoCalSoCal for Sun-as-a-star observations. Echelle spectrographs record light in multiple diffraction orders, each spanning a narrow wavelength range on the detector. Diffraction order 95 was selected because it contains the Fe I lines central to this analysis (near 6388–6390  $\text{cm}^{-1}$ , 1.56  $\mu\text{m}$ ) and is relatively free of telluric contamination, although it is difficult to avoid telluric features entirely in the NIR range.

The analysis began with careful data selection. SoCalSoCal started collecting spectra in October 2024, but data taken from October 1, 2024 to December 16, 2024 suffered from poor quality and irregular sampling due to instrument issues. Data acquired between January–May 2025 were free of instrumental problems. Still, they contained large, unavoidable gaps caused by winter conditions—primarily persistent cloud cover and rain that prevented observations on many days. Because this project focused on periodogram analyses, the analysis was limited to a time span with relatively continuous sampling to reduce artifacts introduced by the observing schedule and sampling pattern. Irregular sampling and gaps distort periodograms by redistributing power into additional peaks and side-peaks set by the spacing between observations, which can create spurious peaks and can smear or broaden true signals. To minimize this effect, the time baseline was restricted to a more continuous subset of observations from June 14, 2025 to August 16, 2025. The time span corresponds to approximately 63 days, which is  $\approx 2.3$  synodic solar equatorial rotations. Because the Sun exhibits differential rotation, magnetic features at higher solar latitudes rotate more slowly, so the effective rotation timescale in disk-integrated data can shift towards longer periods. For instance, a full magnetic rotation takes  $\sim 28$ – $29$  days near the solar equator and  $\sim 35$  days near the poles.

### Detector Calibrations and Order Extraction

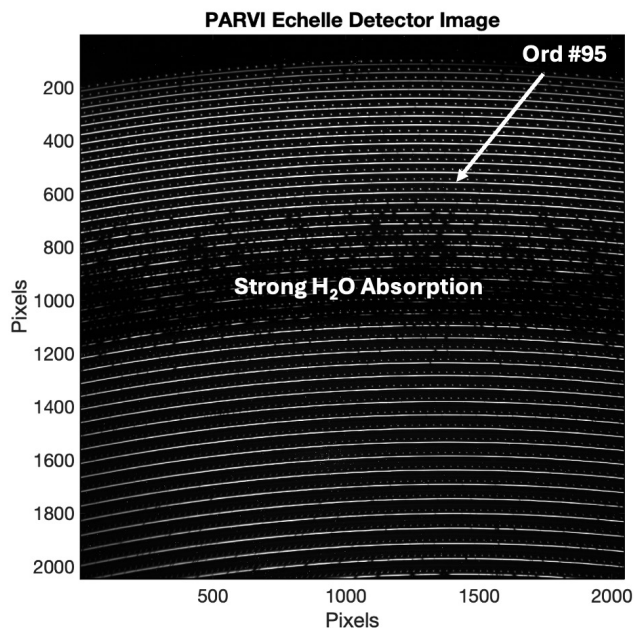
All raw Flexible Image Transport System (FITS) frames were processed in MATLAB with a set of reduction scripts. This MATLAB-based processing workflow applies the calibration and extraction steps described below. Each exposure was cleaned and re-oriented, and it was then corrected for detector background by subtracting a master dark frame. In an echelle spectrograph, the spectrum is dispersed into many “orders,” which appear on the detector as curved, stripe-like traces at different wavelength ranges (Figure 3). The curved echelle orders were traced using high signal-to-noise (SNR) calibration frames, and a smooth trace was fit for each order so that the same detector pixels were extracted consistently from frame to frame. One-dimensional spectra were extracted by summing the signal across each fiber’s order profile, producing both a solar and etalon spectrum for each exposure. Isolated anomalous pixels (hot pixels) were corrected by replacing them with values interpolated from neighboring pixels. The spectra were then sorted by observation time using FITS header timestamps, followed by background-subtraction and normalization to correct for the instrument’s wavelength-dependent sensitivity. The final products were

time-stamped, one-dimensional spectra for both fibers, with the etalon serving as a stable reference to monitor instrument stability.

Since observing conditions varied due to factors such as intermittent clouds, changes in atmospheric seeing, and small guiding or tracking errors that shift the image relative to the fiber, not all spectra were suitable for analysis. For each day of observation, a flux proxy from order 95 was inspected, and low-flux exposures were removed. These removals include short, sudden drops in the flux consistent with passing clouds, as well as the lowest-flux exposures near sunrise and sunset, which is when sunlight passes through the greatest amount of atmosphere, causing the throughput to change quickly. Additionally, exposures flagged in the observing logs due to disk vignetting caused by misalignment of the solar telescope were excluded. Disk vignetting is the partial blocking of the solar disk at the instrument entrance, meaning only part of the sun is recorded. This changes the disk-integrated line profiles in a way that is instrumental rather than solar in origin, and can introduce spurious radial-velocity-like signals. A conservative cutoff was used because the spectral modeling step is sensitive to low-SNR data. All removed exposures were recorded as an exclusion mask. After filtering, all spectra were sorted chronologically by their mid-exposure timestamps from the FITS headers and converted into Modified Julian Date (MJD), a standard astronomy timestamp that counts days continuously, resulting in a time-ordered dataset of about  $10^4$  spectra suitable for subsequent analysis. Within this interval, duplicate timestamps were removed, and spectra with Not-a-Number (NaN) values or corrupted values were discarded.

### Telluric Modeling with Stellar-GFIT

Atmospheric absorption lines from water vapor ( $\text{H}_2\text{O}$ ), deuterated water ( $\text{HDO}$ ), and other molecules were modeled using Stellar-GFIT, a



**Figure 3: Representative PARVI echelle detector image in the near-infrared.** The spectrum is dispersed into multiple diffraction orders. Order 95, which contains the lines analyzed in this study, is indicated by the arrow. The prominent dark band corresponds to strong telluric  $\text{H}_2\text{O}$  absorption. Wavelength increases with left to right within an order, and from bottom to top across the orders.

forward-modeling tool. Running GFIT requires site-specific atmospheric state inputs. These were generated by preparing Palomar Observatory configuration files (site code pr, latitude 33.357°, longitude -116.864°, altitude 1.7 km) and uploading them to JPL's Global Navigation Satellite System (GNSS)-based meteorological data service. The returned state files were linked to the solar spectra.

Each observing run required creation of a .grl (runlog) file in MATLAB, which contains instrumental metadata, observational geometry, and barycentric velocity corrections that adjust for Earth's motion around the Sun. Barycentric corrections were computed separately with the barycorrpy package in Python and appended as additional columns. The .grl file was then registered with GFIT through a configuration utility (gsetup), which generated .ggg input files. After minor header adjustments, GFIT was executed to produce a best-fit telluric model (including H<sub>2</sub>O and HDO) for each exposure in the region of interest (**Figure 4**). In MATLAB, the telluric correction was applied by dividing the observed solar spectrum by the modeled telluric transmission, yielding telluric-corrected spectra used for Gaussian fitting and equivalent-width integration.

### Gaussian Line Fitting

For activity-sensitive diagnostics, two Fe I absorption features in order 95 were targeted. The Fe I 6388.6 cm<sup>-1</sup> line was modeled with a double-Gaussian profile to account for its Zeeman-sensitive core and nearby solar blends. The Fe feature near 6390 cm<sup>-1</sup> required a triple-Gaussian model to separate overlapping components. Fits were performed on normalized, telluric-corrected spectra using nonlinear least-squares minimization with a local linear baseline term (mx + d) in MATLAB using the fit function. From each fit, line depth, centroid position, and full width at half maximum (FWHM) as time-series indicators of solar activity were extracted.

### Equivalent Width Measurement

In addition to profile fits, EW was computed directly from the continuum-normalized, telluric-corrected spectra. For each target line, a region of interest (ROI) spanning the absorption feature was defined. The EW was calculated as

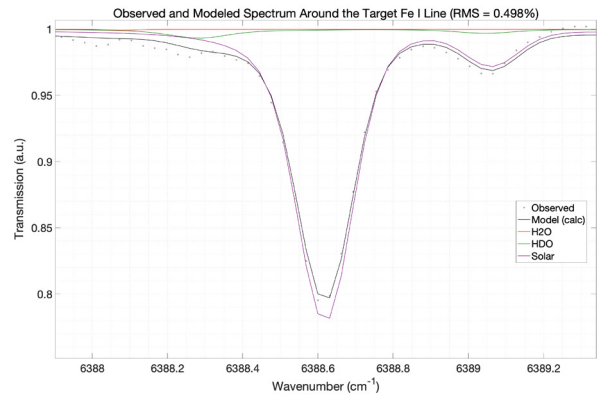
$$EW = \int_{v_1}^{v_2} (1 - F(v)) dv \quad (1)$$

where  $F(v)$  is the continuum-normalized flux as a function of the wavenumber, using trapezoidal numerical integration over the ROI. This provides a measure of line strength less sensitive to assumptions about the detailed line shape. EWs were computed for each exposure, along with timestamps for time-series analysis.

Five diagnostic lines within PARVI's H-band coverage were focused on:

1. Fe I at 6388.6 cm<sup>-1</sup> ( $g_{\text{eff}} = 3$ )
2. Fe I at 6390.05 cm<sup>-1</sup>
3. Fe I at 6399.6 cm<sup>-1</sup>
4. H<sub>2</sub>O at 6401.09 cm<sup>-1</sup> (control line)
5. Ni I at 6406.14 cm<sup>-1</sup>

The first three Fe lines and the Ni line are photospheric features with varying magnetic sensitivities, making them useful markers of solar activity. In contrast, the H<sub>2</sub>O line is of purely telluric origin. Because telluric lines should not vary with intrinsic solar conditions, the H<sub>2</sub>O feature was used as a control; any spurious EW variability would indicate residual atmospheric contamination or reduction artifacts. **Figure 5** shows that the telluric H<sub>2</sub>O control line is strongly sensitive to changing



**Figure 4: Stellar-GFIT model components for the Fe I 6388.6 cm<sup>-1</sup> (1564.8 nm) target line.** Black points show the observed spectrum in the region of interest (ROI), and the black curve shows the Stellar-GFIT best-fit model to the observations. The purple curve shows the telluric-free solar (stellar) model component returned by Stellar-GFIT, included here for reference and not used directly in the downstream equivalent width analysis. The green curve shows the modeled telluric absorption component from deuterated water (HDO). The reported root-mean-square (RMS) residual of 0.498% indicates that the full model reproduces the observed normalized spectrum to within about 0.5% on average across the fitted window. In MATLAB, the observations were corrected by dividing the black spectrum by the full modeled telluric transmission (including HDO and other relevant species) to produce the telluric-corrected spectra used for line fitting and equivalent-width measurements.

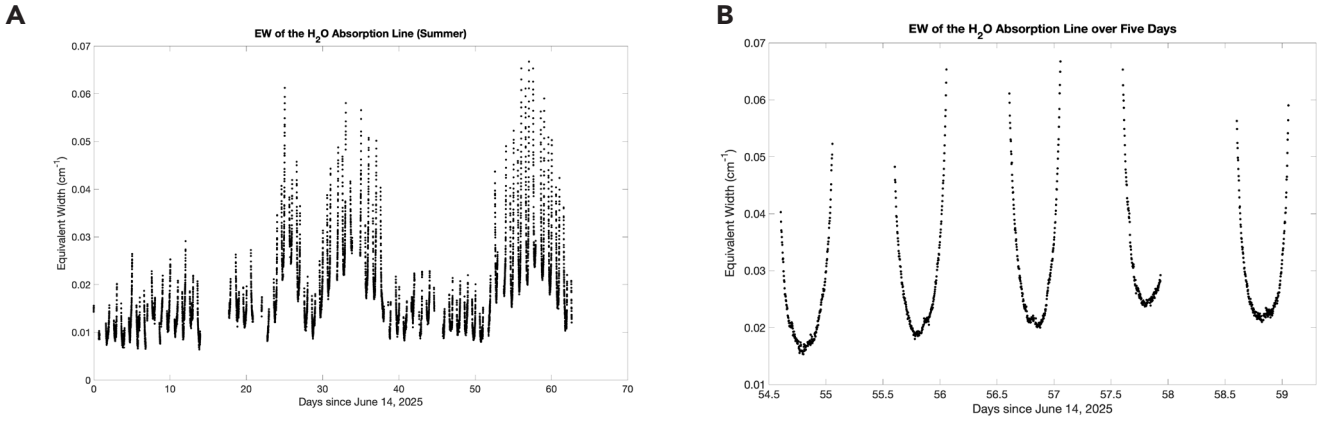
terrestrial observing conditions, particularly on intraday timescales. This figure is included only as a qualitative demonstration of the control line's atmospheric sensitivity. By comparing the Fe and Ni lines against this control, it was ensured that the measured periodicities were of astrophysical rather than terrestrial origin.

EW windows were chosen by eye. This was done by inspecting a high-SNR reference spectrum and defining fixed ROIs for each line that captures each absorption feature while avoiding nearby blends. ROIs were also chosen to avoid wavelengths where telluric correction left noticeable residual structure, which can bias EW measurements. The robustness of the ROI choice was checked by repeating the EW measurements with several slightly shifted ROI boundaries; these tests produced the same dominant periodogram peaks. EWs were integrated in the observed frame (that is, without shifting each spectrum into a common rest frame). Daily line-position drifts are present and likely include barycentric motion.

### Periodogram Computation

Lomb–Scargle periodograms of the EW time series were computed using MATLAB's `plomb`. Series were mean-detrended via `detrend(...,0)` to remove the constant offset so the periodogram emphasizes variability rather than the absolute EW level. Periodograms were searched over periods from 1–65 days; in **Figure 6**, 2–65 days is shown to de-emphasize the strong 1-day sampling peak. A sampling-only spectral window was computed from the observation times to visualize cadence aliases. The upper search limit  $P_{\text{max}} = 65$  d matches the contiguous baseline of June 14–August 16, 2025, avoiding poorly constrained longer periods. False-alarm probabilities (FAP) were estimated using `plomb`'s analytic formulation via the `Pd` option, which returns the critical periodogram level that a peak must exceed to be considered significant at the specified FAP. A fixed 1% false-alarm probability line was not plotted because strong low-frequency periodogram power, the Lomb–Scargle periodogram values returned by `plomb`, can bias the corresponding significance threshold.





**Figure 5: Telluric H<sub>2</sub>O equivalent width shows daily variability.** (A) Equivalent width of a telluric H<sub>2</sub>O line as a function of time, showing diurnal variations and longer-term changes over weeks due to seasonal water vapor. (B) This panel highlights five consecutive days, illustrating pronounced daily fluctuations with airmass.

### Injection-Recovery Benchmark

To validate the Lomb–Scargle periodogram pipeline under the same irregular cadence as the observations and to quantify the impact of the inconsistent sampling window—given the gaps and uneven time spacing of the measurements—on recovered periodicities, an injection test was performed on synthetic EW time series. In an injection test, a known periodic signal is added to a noise-like baseline, and the resulting series is evaluated only at the exact observation timestamps  $t_i$ , and the full periodogram workflow is rerun with identical settings. This benchmark checks whether the pipeline reliably recovers the injected period under the real cadence and reveals how cadence can suppress sensitivity or introduce alias peaks. The recovered peak powers and periods are later used to estimate periodogram noise statistics and uncertainty in recovered signal properties.

10 synthetic EW realizations were generated in the form

$$EW_{\text{syn}}(t_i) = EW_{\text{noise}}(t_i) + A \sin\left(\frac{2\pi t_i}{P} + \phi\right) \quad (2)$$

evaluated only at the observed timestamps  $t_i$ , where  $EW_{\text{noise}}(t_i)$  is a Gaussian random time series, and the injected sinusoid has period  $P = 27.5$  d, fixed semi-amplitude  $A = 6 \times 10^{-5}$  cm<sup>-1</sup>, and phase offset  $\phi$ , which is drawn uniformly from 0 to  $2\pi$  for each realization. For each realization, a Lomb–Scargle periodogram was computed using the same detrending, period limits, frequency grid, and MATLAB implementation (plomb) adopted for the observed EW series.

### Error Calculations

The period corresponding to the strongest peak was refined to sub-bin precision by fitting a parabola to the peak bin and its neighboring bins in the periodogram, with the parabola's vertex taken as the best estimate of the period. Measurement uncertainties were estimated using a Monte Carlo noise-injection approach, wherein synthetic noise consistent with the EW noise in the data and at the observed cadence were generated. For each realization, a Lomb–Scargle periodogram was recomputed using identical settings, and the local background periodogram level was characterized by its mean and standard deviation in a narrow window around the peak. The peak SNR was then calculated as the difference between the peak power and the local mean divided by the local standard deviation, and the period uncertainty was estimated as  $\sigma_P \sim P^2/(T \cdot \text{SNR})$ , where  $P$  is

the peak period determined above and  $T = 63$  d is the total baseline. Similarly, the amplitude uncertainty of each peak can be derived from the scatter in peak heights over the Monte Carlo realizations. This approach provides a statistically motivated estimate of the period and amplitude errors.

## RESULTS

### Photospheric Fe I and Ni I Equivalent-Width Periodograms Show a Consistent Rotation-Timescale Between 27 and 31 Days

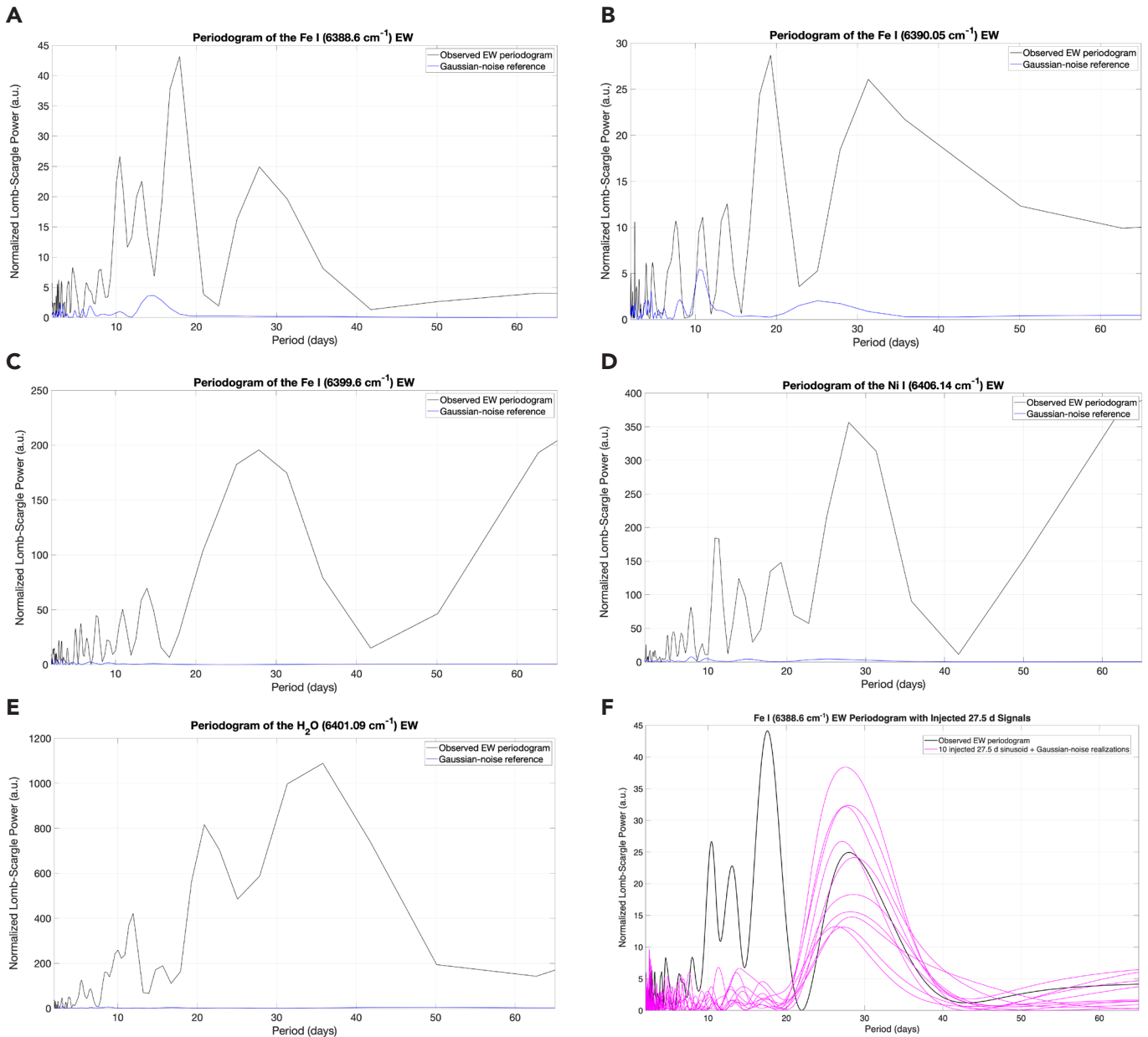
Equivalent widths (EWs) were measured for five diagnostic lines—Fe I 6388.6, 6390.05, 6399.6 cm<sup>-1</sup>, Ni I 6406.14 cm<sup>-1</sup>, and the telluric control H<sub>2</sub>O 6401.09 cm<sup>-1</sup>—over June 14, 2025 to August 16, 2025. Lomb–Scargle periodograms of the photospheric Fe I and Ni I lines show peaks both near 19 d and the rotation-timescale range between 27.5 to 31.5 d (Table 1; Figure 6A–E). Although the approximately 19 d feature is often the higher-power peak, the 27.5 to 31.5 d peak is consistent with the solar synodic rotation period of 27.3 d, making it the feature of primary interest in this analysis. Spectral-window and injection-test comparisons show that the 27.5 to 31.5 d feature is recoverable under the observed cadence (Figure 6F). The telluric H<sub>2</sub>O has a dominant periodogram peak occurring at  $35.5 \pm 1.0$  d does not show a corresponding dominant peak in the 27.5 to 31.5 d range (Table 1; Figure 6E).

### Injection Tests Recover a Peak Near 27.5 D in the Target Fe I Time Series

A 27.5 d period sinusoid with semi-amplitude  $6 \times 10^{-5}$  cm<sup>-1</sup> was added to Gaussian-noise time series evaluated at the exact timestamps of the observed EW measurements (10 realizations). Lomb–Scargle periodograms were then recomputed using the same settings as for the observed EW series. The recovered power clusters near 27.5 d, showing that a rotation-timescale signal is recoverable under the observed cadence and analysis workflow (Figure 6E).

## DISCUSSION

This work examines whether NIR EWs of photospheric lines in disk-integrated solar spectra exhibit periodic variability on rotation timescales, and whether that variability can be separated from residual telluric systematics using a control H<sub>2</sub>O line. Periodogram peaks in the 27 to 31 d range appear in four photospheric lines, whereas the telluric



**Figure 6: Lomb-Scargle periodograms of absorption-line equivalent widths over 2 to 65 days.** Periodograms of three Fe I lines (A to C), including the targeted activity-sensitive Fe I  $6388.6 \text{ cm}^{-1}$  line (A), the Ni I line (D), and the telluric  $\text{H}_2\text{O}$  control line (E). In panels A to E, the blue curves show periodograms of randomized Gaussian-noise EW time series computed using the same observation cadence and periodogram setting, providing a reference for noise- and sampling-driven structure in the absence of an injected periodic signal. (F) shows an injection benchmark for the Fe I  $6388.6 \text{ cm}^{-1}$  line: 10 synthetic EW time series periodograms (magenta), each constructed from Gaussian noise plus a 27.5 day sinusoid with a fixed semi-amplitude of  $6 \times 10^{-5}$ . Unlike the blue noise-only references, the injected magenta realizations tether whether a rotation-timescale signal can be recovered under the actual sampling window; the clustering of their recovered power near 27.5 d is used to assess the detectability and cadence sensitivity of the rotation-period feature.

$\text{H}_2\text{O}$  control does not show a dominant peak in that same interval. This pattern is consistent with the rotation-timescale variability in the photospheric EWs that is distinct from the dominant periodicities in the telluric control. The injection benchmark further indicates that a signal near 27.5 d is recoverable under the observed cadence, which argues against the 27 to 31 d feature arising solely from the sampling pattern. For the Fe I  $6388.6 \text{ cm}^{-1}$  line, the observed periodic EW variability likely reflects a combination of magnetic broadening and splitting, temperature contrasts associated with sunspots and faculae,

and changes in the visible coverage of these regions across the solar disk (3–4, 11). The recurrence of a 28 and 31 d signal across multiple photospheric lines is also consistent with rotational modulation from evolving active regions, which provides context for interpreting this timescale as solar rather than telluric in origin (7–8, 11).

Several factors limit period precision and can introduce secondary peaks. First, the 63-day baseline spans only about two solar rotations, which reduces frequency resolution and naturally broadens rotation-scale peaks. Second, irregular cadence and gaps can introduce window-

function aliases (10). Third, EW measurements can be affected by continuum-placement choices and small line-position drifts when the integration window is fixed in the observed frame. Given the short baseline and broad peaks, these period detections are better interpreted as rotation-consistent timescales than as precise measurements of a single stable period. Variable observing conditions and residual reduction systematics can further broaden peaks and can also produce spurious low-power features elsewhere in the periodograms. These limitations motivate measuring EWs over longer observing baselines in future work and applying more complete uncertainty propagation, including uncertainties associated with continuum fitting, line-window placement, and residual wavelength drift.

| Line                                                | Peak 1 (d) |
|-----------------------------------------------------|------------|
| Fe I 6388.6 cm <sup>-1</sup> (primary)              | 27.8 ± 2.2 |
| Fe I 6390.05 cm <sup>-1</sup>                       | 31.5 ± 3.2 |
| Fe I 6399.6 cm <sup>-1</sup>                        | 27.5 ± 0.3 |
| Ni I 6406.14 cm <sup>-1</sup>                       | 28.9 ± 0.9 |
| H <sub>2</sub> O 6401.09 cm <sup>-1</sup> (control) | 35.5 ± 1.0 |

**Table 1: Rotation-timescale EW periods by line (days).** Peak 1 reports the Lomb–Scargle periodogram peak associated with the rotation-timescale feature near 27.5 to 31.5 for each equivalent-width (EW) time series measured from June 14 to August 16, 2025. Reported values are peak periods in days with  $\pm 1\sigma$  uncertainties derived from peak-shape fitting and noise-based periodogram simulations. Fe I and Ni I are photospheric lines, while H<sub>2</sub>O is a telluric control line used to identify periodicities associated with Earth’s atmosphere and/or residual reduction systematics.

### Implications and Future Directions

If near-infrared EWs respond to solar magnetic and photospheric activity differently from optical lines, then combining both wavelength regimes may improve the separation of activity-driven stellar signals from true Doppler shifts (2). Existing activity-mitigation approaches, including photometric proxies and line-by-line radial velocity analyses, are informative but do not always clearly distinguish wavelength-dependent activity signals from planet-like radial velocity variations. In this context, the present study provides an initial validation that EW monitoring of magnetically sensitive NIR photospheric lines can recover rotation-timescale variability while remaining distinguishable from the dominant periodicities in a telluric control line. This offers a simple and potentially stable complement to more complex diagnostics, particularly for testing whether activity signatures behave differently in the NIR and optical.

A natural next step is to correlate the NIR EW measurements with solar activity tracers derived from the Solar Dynamics Observatory, including daily spot fill factors and magnetograms (11) and with contemporaneous optical Zeeman-sensitive lines observed by visible-light spectrographs such as NEID and HARPS-N (7–8). These comparisons would test for wavelength-dependent, or wavelength-differential, activity responses, meaning differences in how strongly activity signatures appear in the NIR relative to the optical.

Additional future work could include extending the monitoring baseline to test whether the observed patterns persist over longer timescales, analyzing additional NIR lines with different magnetic sensitivities to better distinguish thermal from magnetic effects (3–4), comparing the NIR EW variations to optical tracers and RV residuals (7–8),

and improving uncertainty estimates by incorporating exposure-by-exposure noise and continuum placement tests. Because micro-telluric contamination can bias precision measurements even when individual features are weak, a particularly important validation step is to test the sensitivity of the results to perturbations in the telluric model (6, 12). Together, these steps would clarify how NIR EWs can contribute to multi-wavelength strategies for mitigating stellar jitter and improving the detection of small exoplanets (2).

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